



FYDP-DSE D3

[FPSEF25X05]

***Bachelors of Science in Software Engineering
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1. Final Source Code Link (GitHub)

The complete and final version of the project's source code is hosted in a well-structured GitHub repository. This repository contains all components of the food image analysis pipeline including classification, segmentation, portion estimation, and nutrient profiling modules.

- GitHub Repository: <https://github.com/foodanalyzer/SmartFoodAnalyzer>
- Last Commit; Integrated complete pipeline with all four modules

2. README with Usage Instructions

Project Overview

This project implements a comprehensive computer vision pipeline for automated food analysis. The system takes food images as input and provides detailed nutritional information through a four-stage process:

1. **Food Classification** - Identifies the type of food item
2. **Image Segmentation** - Isolates food items from background
3. **Portion Estimation** - Calculates volume and weight using reference objects
4. **Nutrient Profiling** - Maps food items to nutritional databases for calorie and nutrient calculation

Technology Stack

- **Programming Language:** Python 3.9+
- **Deep Learning Frameworks:** PyTorch 2.0+, TensorFlow 2.12+
- **Computer Vision:** OpenCV 4.8



- **Models:** YOLOv8-seg, Mask-RCNN (Segmentation), ResNet50/MobileNet (Classification),
- **Data Processing:** NumPy, Pandas, Scikit-learn
- **Visualization:** Matplotlib, Seaborn, Plotly
- **Database:** JSON-based nutrient database (USDA compatible)

Key Features

- **Multi-class Food Classification** (85 food categories)
- **Instance Segmentation** with pixel-level accuracy
- **Portion Size Estimation** using reference object scaling
- **Comprehensive Nutrient Profiling** (calories, protein, carbs, fats)
- **Batch Processing** capability for multiple images
- **Export Results** in JSON, CSV, and PDF formats
- **Accuracy Metrics** generation for model evaluation

System Requirements

- **Minimum:** Python 3.9, 8GB RAM, 2GB GPU VRAM (for CUDA acceleration)
- **Recommended:** Python 3.11, 16GB RAM, 8GB GPU VRAM (NVIDIA GPU with CUDA 11).
- **Storage:** 10GB free space for models and datasets

Installation & Setup



Quick Installation

```
git clone https://github.com/foodanalyzer/SmartFoodAnalyzer
cd SmaerFoodAnalyzer

# Create a virtual environment
python -m venv venv
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt
```

3. Setup Manual for Cross-Platform Installation

This section provides detailed installation instructions for all supported platforms to ensure consistent performance across different environments.

Prerequisites

- Python 3.9 or higher
- Git
- Minimum 8GB RAM
- CUDA-capable GPU (optional, for faster inference)

Windows Installation

Step 1: Install Python

1. Download Python 3.9+ from python.org
2. During installation, check "Add Python to PATH"
3. Verify installation:

```
python --version
pip --version
```



Step 2: Install Git

1. Download Git from git-scm.com
2. Use Git Bash for command-line operations

Step 3: Clone Repository

```
git clone https://github.com/foodanalyzer/SmartFoodAnalyzer
cd SmaerFoodAnalyzer
```

Step 4: Create Virtual Environment

```
python -m venv nutri_env
nutri_env\Scripts\activate
```

Step 5: Install Dependencies

```
pip install --upgrade pip
pip install -r requirements.txt
```

Step 6: Install PyTorch with CUDA (for GPU support)

```
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118
```

Step 7: Download Models

```
python download_models.py
```